

- 361.** Marks obtain by all the students of class 12 are presented in a frequency distribution with classes of equal width. Let the median of this grouped data be 14 with median class interval 12-18 and median class frequency 12 . If the number of students whose marks are less than 12 is 18 , then the total number of students is :

कक्षा 12 के सभी छात्रों द्वारा प्राप्त प्राप्तांकों को एक बारम्बारता बंटन में समान अन्तराल के वर्गों के साथ प्रदर्शित किया जाता है। माना कि माधिका वर्गान्तराल 12-18 और माधिका वर्ग बारम्बारता 12 के साथ, इन वर्गीकृत आँकड़ों की माधिका 14 है। यदि 12 से कम प्राप्तांकों वाले छात्रों की संख्या 18 है, तो छात्रों की कुल संख्या है :

- (A) 52 (B) 44 (C) 40 (D) 48

Ans. B

Sol.
$$M = \ell + \left(\frac{\left(\frac{N}{2} - C \right)}{f} \right) h$$

$$\ell = 12$$

$$C = 18$$

$$M = 14$$

$$f = 12$$

$$h = 6$$

$$N = ?$$

$$M = 12 + \left(\frac{\frac{N}{2} - 18}{12} \right) 6 = 14$$

$$\frac{\frac{N}{2} - 18}{2} = 2$$

$$\frac{N}{2} = 22$$

$$N = 44$$

- 362.** For a statistical data $x_1, x_2, \dots, x_{10}$ of 10 values, a student obtained the mean as 5.5 and $\sum_{i=1}^{10} x_i^2 = 371$. He later found that he had noted two values in the data incorrectly as 4 and 5, instead of the correct values 6 and 8, respectively. The variance of the corrected data is :

10 मानों के सांख्यिकीय आँकड़ों $x_1, x_2, \dots, x_{10}$ के लिए एक छात्र ने माध्य 5.5 प्राप्त किया और $\sum_{i=1}^{10} x_i^2 = 371$ है। उसने बाद में पाया कि उसने सही आँकड़ों में सही मानों 6 और 8 के स्थान पर गलत मानों क्रमशः 4 और 5 को लिख लिया था। सही किए गए आँकड़ों का प्रसरण है :

(A) 9

(B) 4

(C) 7

(D) 5

Ans. C

Sol. $x_1 + x_2 + \dots + x_8 + 4 + 5 = 55 \quad \dots(1)$

$x_1^2 + x_2^2 + \dots + x_8^2 + 4^2 + 5^2 = 371 \quad \dots(2)$

For corrected data mean is

$$\frac{x_1 + x_2 + \dots + x_8 + 6 + 8}{10} = \frac{55 - 9 + 6 + 8}{10} = 6$$

$$\sigma^2 + x^{-2} = \frac{1}{n} \sum x_i^2$$

$$\sigma^2 + 6^2 = \frac{1}{10} (371 - 4^2 - 5^2 + 36 + 64)$$

$$= \frac{1}{10} (471 - 41) = \frac{430}{10}$$

$$= \sigma^2 = 7$$

363. Let $x_1, x_2, \dots, x_{10}$ be ten observations such that $\sum_{i=1}^{10} (x_i - 2) = 30$, $\sum_{i=1}^{10} (x_i - \beta)^2 = 98$, $\beta > 2$ and their variance is $\frac{4}{5}$.

If μ and σ^2 are respectively the mean and the variance of $2(x_1 - 1) + 4\beta, 2(x_2 - 1) + 4\beta, \dots, 2(x_{10} - 1) + 4\beta$, then

$\frac{\beta\mu}{\sigma^2}$ is equal :

माना कि $x_1, x_2, \dots, x_{10}$ दस प्रेक्षण इस प्रकार हैं कि $\sum_{i=1}^{10} (x_i - 2) = 30$, $\sum_{i=1}^{10} (x_i - \beta)^2 = 98$, $\beta > 2$ और उनका प्रसरण $\frac{4}{5}$ है।

$2(x_1 - 1) + 4\beta, 2(x_2 - 1) + 4\beta, \dots, 2(x_{10} - 1) + 4\beta$ क माध्य और प्रसरण क्रमशः यदि μ और σ^2 हैं, तो $\frac{\beta\mu}{\sigma^2}$ बराबर है :

(A) 110

(B) 90

(C) 120

(D) 100

Ans. D

Sol. $\sum_{i=1}^{10} (x_i - 2) = 30$

$$\sum x_i - 2 \sum 1 = 30$$

$$\sum x_i - 20 = 30$$

$$\sum x_i = 50$$

$$\sum (x_i - \beta)^2 = 98$$

$$\sum (x_i^2 + \beta^2 - 2\beta x_i) = 98$$

$$\sum x_i^2 + \beta^2 \sum 1 - 2\beta \sum x_i = 98$$

$$\sum x_i^2 + 10\beta^2 - 2\beta \times 50 = 98$$

$$\sum x_i^2 = 98 - 10\beta^2 + 100\beta$$

$$\sigma^2 = \frac{\sum x_i^2}{10} - \left(\frac{\sum x_i}{10} \right)^2 = \frac{4}{5}$$

$$\Rightarrow \frac{98 - 10\beta^2 + 100\beta}{10} - \left(\frac{50}{10} \right)^2 = \frac{4}{5}$$

$$\beta^2 - 10\beta + 16 = 0$$

$$\beta = 2, 8$$

$$\beta = 8$$

$$2x_1 - 2 + 4\beta, 2x_2 - 2 + 4\beta, \dots$$

$$2x_1 + 30, 2x_2 + 30, 2x_3 + 30, \dots, 2x_{10} + 30$$

$$\mu = \frac{\sum (2x_i + 30)}{10} = \frac{2\sum x_i}{10} + 30$$

$$= 2 \times 5 + 30 = 40$$

$$\sigma^2 = \frac{\sum (2x_i + 30)^2}{10} - (40)^2$$

$$\sigma^2 = \frac{16}{5}$$

$$\text{Now, } \frac{\beta\mu}{\sigma^2} = 100$$

364. The frequency distribution of the age of students in a class of 40 students is given below

Age	15	16	17	18	19	20
No of Students	5	8	5	12	x	y

If the mean deviation about the median is 1.25, then $4x + 5y$ is equal to :

40 छात्रों को एक कक्षा के छात्रों की आयु का बारंबारता बंटन नीचे दिया है

आयु	15	16	17	18	19	20
छात्रों की संख्या	5	8	5	12	x	y

यदि माध्यिका के सापेक्ष माध्य विचलन 1.25 है, तो $4x + 5y$ बराबर है :

(A) 43

(B) 46

(C) 44

(D) 47

Ans. C

Sol.

Age	No. of student	C.f	$ x_c - 18 $	$f_c x_c - 18 $
15	5	5	3	15
16	8	13	2	16
17	5	18	1	5
18	12	30	0	0
19	x	30 + x	1	x
20	y	30 + x + y	2	2y

$$N = \sum f_i = 30 + x + y = 40$$

$$x + y = 10 \quad \dots\dots\dots(1)$$

$$\frac{N}{2} = 20$$

So, median of given observers is 18

$$\sum f_i |x_i - 18| = 36 + x + 2y$$

$$\text{Mean deviation about median} = \frac{\sum f_i |x_i - 18|}{\sum f_i}$$

$$\Rightarrow \frac{36 + x + 2y}{40} = 1.25$$

$$\Rightarrow 36 + x + 2y = 50$$

$$\Rightarrow x + 2y = 14 \quad \dots\dots\dots(2)$$

$$\text{So, } y = 4, x = 6$$

$$4x + 5y = 24 + 20 = 44$$

365. If the variance of the frequency distribution

x	c	$2c$	$3c$	$4c$	$5c$	$6c$
f	2	1	1	1	1	1

is 160, then the value of $c \in \mathbb{N}$ is :

यदि बारंबारता बंटन

x	c	$2c$	$3c$	$4c$	$5c$	$6c$
f	2	1	1	1	1	1

का प्रसरण 160 है, तो $c \in \mathbb{N}$ का मान है :

- (A) 6 (B) 7 (C) 8 (D) 5

Ans. B

Sol. $\sigma^2 = \frac{\sum f_i x_i^2}{\sum f_i} - \left(\frac{\sum f_i x_i}{\sum f_i} \right)^2 = 160$

$$\Rightarrow \frac{c^2(2+4+9+16+25+36)}{7} - \left(\frac{c \cdot (2+2+3+4+5+6)}{7} \right)^2 = 160$$

$$c^2 \times \frac{92}{7} - c^2 \times \left(\frac{22}{7} \right)^2 = 160$$

$$\frac{c^2}{49} \times \{92 \times 7 - 4 \times 121\} = 160$$

$$\Rightarrow \frac{c^2}{49} \times 160 = 160$$

$$\Rightarrow c = 7$$

366. Let $a_1, a_2, \dots, a_{10}$ be 10 observations such that $\sum_{k=1}^{10} a_k = 50$ and $\sum_{\forall k < j} a_k \cdot a_j = 1100$. Then the standard deviation of $a_1, a_2, \dots, a_{10}$ is equal to :

माना 10 प्रेक्षणों $a_1, a_2, \dots, a_{10}$ के लिए $\sum_{k=1}^{10} a_k = 50$ तथा $\sum_{\forall k < j} a_k \cdot a_j = 1100$ हैं। तो $a_1, a_2, \dots, a_{10}$ का मानक विचलन बराबर है :

(A) $\sqrt{115}$ (B) 10 (C) $\sqrt{5}$ (D) 5

Ans. C

Sol. $\sum_{k=1}^{10} a_k = 50 \Rightarrow \mu = 5$

$$\text{variance} = \frac{\sum (a_i)^2}{n} - (\mu)^2 \quad \underline{\hspace{2cm}} (1)$$

$$(a_1 + a_2 + \dots + a_{10})^2 = \sum_{\substack{i \neq j \\ i, j=1,2,\dots,10}} (a_i^2 + 2(a_i a_j))$$

$$\Rightarrow (50)^2 - 2(1100) = \sum a_i^2$$

$$\Rightarrow \sum a_i^2 = 2500 - 2200$$

$$\Rightarrow \sum a_i^2 = 300$$

$$\text{variance} = \frac{300}{10} - 25$$

$$\text{variance} = \frac{50}{10} = 5$$

$$\text{S.D.} = \sqrt{5}$$

- 367.** If the mean and variance of five observations are $\frac{24}{5}$ and $\frac{194}{25}$ respectively and the mean of the first four observations is $\frac{7}{2}$, then the variance of the first four observations is equal to :

यदि पाँच प्रेक्षणों के माध्य तथा प्रसरण क्रमशः $\frac{24}{5}$ तथा $\frac{194}{25}$ हैं, प्रथम चार प्रेक्षणों का माध्य $\frac{7}{2}$ है, तो प्रथम चार प्रेक्षणों का प्रसरण बराबर है :

- (A) $\frac{77}{12}$ (B) $\frac{5}{4}$ (C) $\frac{4}{5}$ (D) $\frac{105}{4}$

Ans. B

Sol.
$$\frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = \frac{24}{5}$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 + x_5 = 24$$

$$\frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{7}{2}$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 = 14$$

$$\therefore x_5 = 10$$

also
$$\frac{x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2}{5} - \frac{576}{25} = \frac{194}{25}$$

$$\Rightarrow x_1^2 + x_2^2 + x_3^2 + x_4^2 = 54$$

$$\therefore \text{Var.} = \frac{\sum_{i=1}^4 x_i^2}{4} - \left(\frac{\sum x_i}{4} \right)^2$$

$$= \frac{54}{4} - \frac{49}{4} = \frac{5}{4}$$

368. Let M denote the median of the following frequency distribution.

Class	0 – 4	4 – 8	8 – 12	12 – 16	16 – 20
Frequency	3	9	10	8	6

Then 20M is equal to :

माना बारम्बारता बंटन

वर्ग	0 – 4	4 – 8	8 – 12	12 – 16	16 – 20
बारम्बारता	3	9	10	8	6

की माध्यिका M है। तो 20M बराबर है :

(A) 416 (B) 208 (C) 104 (D) 52

Ans. B

Sol.

Class	Frequency	Cumulative frequency
0 - 4	3	3
4 - 8	9	12
8 - 12	10	22
12 - 16	8	30
16 - 20	6	36

$$M = l + \left(\frac{\frac{N}{2} - C}{f} \right) h$$

$$M = 8 + \frac{18-12}{10} \times 4$$

$$M = 10.4$$

$$20M = 208$$

- 369.** Let the median and the mean deviation about the median of 7 observation 170, 125, 230, 190, 210, a, b be 170 and $\frac{205}{7}$ respectively. Then the mean deviation about the mean of these 7 observations is :

माना 7 प्रेक्षणों 170, 125, 230, 190, 210, a, b के माध्यिका तथा माध्यिका के सापेक्ष माध्य विचलन क्रमशः 170 तथा $\frac{205}{7}$ हैं। तो

इन 7 प्रेक्षणों का माध्य के सापेक्ष माध्य विचलन है :

- (A) 31 (B) 28 (C) 30 (D) 32

Ans. C

Sol. 125, a, b, 170, 190, 210, 230

$$\sum \frac{|x_i - \bar{x}|}{n} = \text{mean deviation about mean}$$

$$\frac{170 - 125 + 170 - a + 170 - b + 190 - 170 + 210 - 170 + 230 - 170}{7} = \frac{205}{7}$$

$$45 + 340 - a - b + 20 + 40 + 60 = 205$$

$$a + b = 300$$

$$\bar{x} = \frac{a + b + 125 + 170 + 190 + 210 + 230}{7} = \frac{1225}{7} = 175$$

Mean deviation about mean =

$$\frac{175 - 125 + 175 - a + 175 - b + 175 - 170 + 190 - 175 + 210 - 175 + 230 - 175}{7}$$

$$= \frac{50 + 350 - 300 + 5 + 15 + 35 + 55}{7} = 30$$

- 370.** Consider 10 observations $x_1, x_2, \dots, x_{10}$ such that $\sum_{i=1}^{10} (x_i - \alpha) = 2$ and $\sum_{i=1}^{10} (x_i - \beta)^2 = 40$, where α, β are positive integers. Let the mean and the variance of the observations be $\frac{6}{5}$ and $\frac{84}{25}$ respectively. Then $\frac{\beta}{\alpha}$ is equal to :

10 प्रेक्षणों $x_1, x_2, \dots, x_{10}$ के लिए $\sum_{i=1}^{10} (x_i - \alpha) = 2$ तथा $\sum_{i=1}^{10} (x_i - \beta)^2 = 40$ हैं, जहाँ α तथा β धनात्मक पूर्णांक हैं। माना इन

प्रेक्षणों के माध्य तथा प्रसरण क्रमशः $\frac{6}{5}$ तथा $\frac{84}{25}$ हैं। तो $\frac{\beta}{\alpha}$ बराबर है :

- (A) $\frac{5}{2}$ (B) 2 (C) 1 (D) $\frac{3}{2}$

Ans. B

Sol. $\frac{\sum x_i}{10} = \frac{6}{5}$ $\sum x_i = 12$ $\sum (x_i - \alpha) = 2$ $\sum x_i - 10\alpha = 2$ $12 - 10\alpha = 2$ $10\alpha = 10$ $\Rightarrow \alpha = 1$ $\Rightarrow \frac{\beta}{\alpha} = 2$	$\frac{\sum x_i^2}{10} - \left(\frac{6}{5}\right)^2 = \frac{84}{25}$ $\sum x_i^2 = 48$ $\sum (x_i^2 - 2x_i\beta + \beta^2) = 40$ $\sum x_i^2 - 2\beta \sum x_i + 10\beta^2 = 40$ $48 - 2\beta(12) + 10\beta^2 = 40$ $10\beta^2 - 24\beta + 8 = 0$ $5\beta^2 - 12\beta + 4 = 0$ $(\beta - 2)(5\beta - 2) = 0$ $\beta = 2 \quad (\because \beta \in \mathbb{N})$
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371. The variance of the numbers 8, 21, 34, 47, ..., 320 is _____.

संख्याओं 8, 21, 34, 47, ..., 320 का प्रसरण है _____.

Ans. 8788

Sol. $8 + (n - 1)13 = 320$

$$13n = 325$$

$$n = 25$$

$$\mu = \frac{\sum x_i}{n} = \frac{8 + 21 + \dots + 320}{25}$$

$$= \frac{\frac{25}{2}(8 + 320)}{25}$$

$$= 164$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - \mu^2$$

$$= \frac{8^2 + 21^2 + \dots + 320^2}{25} - 164^2$$

$$= 8788$$

372. Let $a, b, c \in \mathbb{N}$ and $a < b < c$. Let the mean, the mean deviation about the mean and the variance of the 5 observations 9, 25, a, b, c be 18, 4 and $\frac{136}{5}$, respectively. Then $2a + b - c$ is equal to _____.

माना $a, b, c \in \mathbb{N}$ हैं तथा $a < b < c$ हैं। माना 5 प्रेक्षणों 9, 25, a, b, c का माध्य, माध्य के सापेक्ष माध्य विचलन तथा प्रसरण क्रमशः

18, 4 तथा $\frac{136}{5}$ हैं। तो $2a + b - c$ बराबर है _____.

Ans. 33

Sol. $a, b, c \in \mathbb{N}$ $a < b < c$

$$\bar{x} = \text{mean} = \frac{9 + 25 + a + b + c}{5} = 18$$

$$a + b + c = 56$$

$$\text{Mean deviation} = \frac{\sum |x_i - \bar{x}|}{n} = 4$$

$$= 9 + 7 + |18 - a| + |18 - b| + |18 - c| = 20$$

$$= |18 - a| + |18 - b| + |18 - c| = 4$$

$$\text{Variance} = \frac{\sum |x_i - \bar{x}|^2}{n} = \frac{136}{5}$$

$$= 81 + 49 + |18 - a|^2 + |18 - b|^2 + |18 - c|^2 = 136$$

$$= (18 - a)^2 + (18 - b)^2 + (18 - c)^2 = 6$$

$$\text{Possible values } (18 - a)^2 = 1, (18 - b)^2 = 1, (18 - c)^2 = 4$$

$$a < b < c$$

$$\text{so} \quad 18 - a = 1 \quad 18 - b = -1 \quad 18 - c = -2$$

$$a = 17 \quad b = 19 \quad c = 20$$

$$a + b + c = 56$$

$$2a + b - c = 34 + 19 - 20 = 33$$